

CS & IT ENGINEERING



Computer Network

MAC Sub-Layer

Lecture No. - 06

By - Abhishek Sir





Recap of Previous Lecture



Topic

CSMA/CD ✓

Topic

Ethernet



Topics to be Covered



Topic

Ethernet

Topic

Back-off Time

Topic

Ethernet Bridging

ABOUT ME



Hello, I'm **Abhishek**

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#Q. A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

[GATE 2004]

- ✓ (A) 10000 bits
- (B) 10000 bytes
- (C) 5000 bits
- (D) 5000 bytes

Ans: A

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (1 \text{ km} / 2 \times 10^8 \text{ m/sec}) * 1 \text{ Gbps} \\
 &= 2 * (10^3 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 10^9 \text{ bits/sec} \\
 &= 10000 \text{ bits} \\
 &= 1250 \text{ bytes}
 \end{aligned}$$

#Q. The minimum frame size required for a CSMA/CD based computer network running at 1 Gbps on a 200 m cable with a link speed of 2×10^8 m/s is :

[GATE 2008]

(A) 125 bytes

✓ (B) 250 bytes

(C) 500 bytes

(D) None of these

Ans: B

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (200 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 1 \text{ Gbps} \\
 &= 2 * (200 \text{ m} / 2 \times 10^8 \text{ m/sec}) * 10^9 \text{ bits/sec} \\
 &= 2000 \text{ bits} \\
 &= \boxed{250 \text{ bytes}}
 \end{aligned}$$

#Q. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is bytes.

[GATE 2016]

Ans = 200

Solution :

$$\text{Minimum frame size} = 2 * t_p * \text{Bandwidth}$$

$$= 2 * 40 \text{ microsec} * 20 * 10^6 \text{ bits / second}$$

$$= 2 * 40 * 10^{-6} \text{ second} * 20 * 10^6 \text{ bits / second}$$

$$= 1600 \text{ bits}$$

$$= \boxed{200 \text{ bytes}}$$

#Q. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

[GATE 2015]

(A) 8000

(B) 10000

(C) 16000

✓ (D) 20000

Ans. D

Solution :

$$t_x = 2 t_p$$

$$\text{Minimum Signal Speed} = 2 * \text{Distance} * (\text{Bandwidth} / \text{Frame size})$$

$$= 2 * 1 \text{ km} * (100 \text{ Mbps} / 1250 \text{ bytes})$$

$$= 2 * 1 \text{ km} * (10^8 \text{ bits per second} / 10^4 \text{ bits})$$

$$= 20000 \text{ km per second}$$

[CSMA/CD]

#Q. Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (*in bits*) required for collision detection is _____.

[GATE-2024, Set-2, 2-Mark]

Ans = 500

Solution :

$$\begin{aligned}
 \text{Minimum frame size} &= 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth} \\
 &= 2 * (500 \text{ meters} / 2 \times 10^8 \text{ meters/sec}) * 10^8 \text{ bits/sec} \\
 &= \boxed{500 \text{ bits}}
 \end{aligned}$$

#Q. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

[GATE 2013]

(A) 1

✓ (B) 2

(C) 2.5

(D) 5

Ans: B

$$t_x = 2 t_p$$

Solution :

$$\text{Maximum Distance} = (\text{Frame size} / \text{Bandwidth}) * (\text{Signal Speed} / 2)$$

$$= (10000 \text{ bits} / 500 \text{ Mbps}) * (200000 \text{ km/sec}) / 2$$

$$= (10^4 \text{ bits} / 5 * 10^8 \text{ bits/sec}) * (2 * 10^5 \text{ km/sec}) / 2$$

$$= \boxed{2 \text{ km}}$$



Topic : Ethernet



- Minimum frame size : 64 Bytes
- Maximum frame size : 1518 Bytes

- Inter-frame gap between frames
- Ethernet uses Jamming Signal

MTU for
Ethernet
is 1500 byte

[MTU is max^m
payload of data link
layer frame]

IP Packet size
(including Header) $\leq$ MTU



Topic : Ethernet Frame Format

IEEE 802.3



[Frame]

Preamble	Destination MAC Address	Source MAC Address	<u>Type</u>	Payload	<u>FCS</u> (CRC)
----------	----------------------------	-----------------------	-------------	---------	---------------------

Additional
to the frame
(added by
physical layer)

Header
(14 byte)

[46 byte
to
1500 byte]

Footer
(4 byte)



Topic : Preamble

⇒ Ethernet [8 byte] [No EFD]



- 7 byte binary string of alternate (1 and 0)
- Allowing receivers to synchronize their clock at the bit-level with the transmitter clock.
- Preamble is followed by the one byte SFD
- SFD ends with a 1 instead of 0
[To break the bit pattern of the preamble and signal the start of the actual frame]

IEEE 802.3
7 byte preamble
one byte SFD
one byte EFD



Topic : Type



- Specifies the protocol of the payload
[e.g IPv4, IPv6, ARP etc]
- if [Type < 46 :]
then Type is size of data in payload field bytes
- Size of padding = [46 - Type] bytes

Type

- ① Size of Data in payload in case of padding
- ② Protocol Type



Topic : FCS



→ Frame Check Sequence (FCS)

→ 32 bits (4 bytes)

→ CRC - 32

$$G(x) = x^{32} + \dots + 1$$



Topic : Exponential Backoff Algorithm



-> Binary Exponential Backoff Algorithm

-> At k^{th} collision of particular frame : $[k = 1, 2, 3, \dots]$

if $k < 15$

then transmitter chooses a number R randomly
in between $[0 \text{ to } (2^i - 1)]$ where $[i = \min(k, 10)]$

else

Abort the retransmission, inform to higher layer

if $k \leq 10$
 $i = k$
if $k > 10$
 $i = 10$



Topic : Exponential Backoff Algorithm



-> Wait Time = $\left[R * t_x \right]$

-> Transmitter will sense the channel after Wait Time
[for retransmission of the frame]



Topic : Exponential Backoff Algorithm

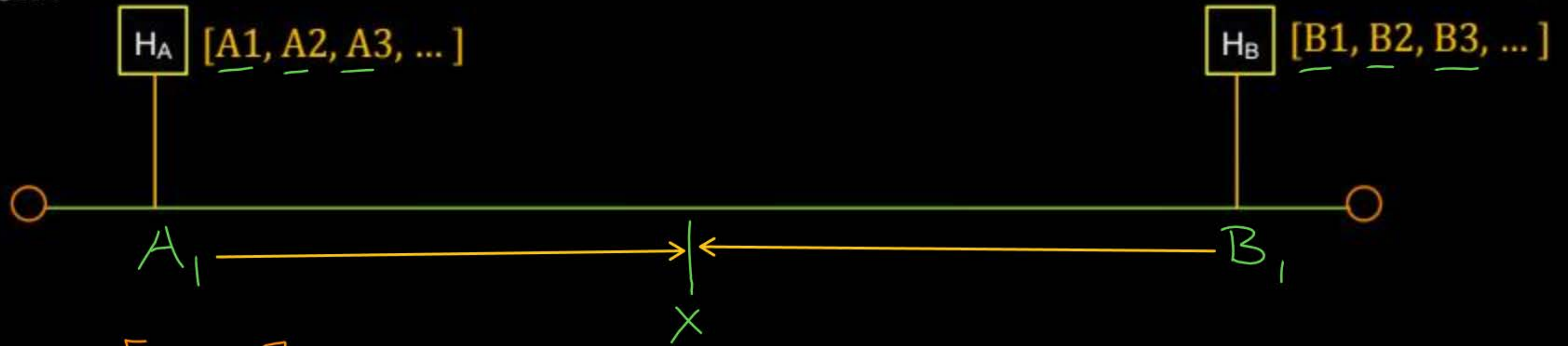


k^{th} collision	Range
1 st	0, 1
2 nd	0, 1, 2, 3
3 rd	0, 1, 2, 3, 4, 5, 6, 7
4 th	0 - - - - - 15
⋮	
9 th	0 - - - - - 511
10 th	0 - - - - - 1023
11 th	0 - - - - - 1023
14 th	0 - - - - - 1023

Example 6 :-

Consider A and B are the only two stations on an Ethernet LAN. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame and collide, what is the probability that A wins the first backoff race ?

Solution :-



$$A_1 \rightarrow [0, 1]$$

$$B_1 \rightarrow [0, 1]$$

A1	B1	Result
0	0	Collied Again
1	1	Collied Again
0	1	A win ✓
1	0	B win

$$\boxed{\text{Ans} = \frac{1}{4}}$$

#Q. A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is:

- (A) 0.5
- (B) 0.625
- (C) 0.75
- (D) 1.0

[GATE 2004]

Handwritten notes in orange ink:
 IIT-D
 H.W.

#Q. In an Ethernet local area network, which one of the following statements is TRUE?

[GATE 2016]

ISC, H.W.

- (A) A station stops to sense the channel once it starts transmitting a frame.
- (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
- (C) A station continues to transmit the packet even after the collision is detected.
- (D) The exponential backoff mechanism reduces the probability of collision on retransmissions.



Topic : LAN Standards



→ IEEE 802.3 : Ethernet [Bus Topology => CSMA/CD]

→ IEEE 802.4 : Token Bus [Bus Topology => Token]

→ IEEE 802.5 : Token Ring [Ring Topology => Token]

→ IEEE 802.11 : Wireless [Wi-Fi => CSMA/CA]

#Q. Which of the following statements is TRUE about CSMA/CD?

[GATE 2005]

IIT-B
H.W.

- (A) IEEE 802.11 wireless LAN runs CSMA/CD protocol.
- (B) Ethernet is not based on CSMA/CD protocol.
- (C) CSMA/CD is not suitable for a high propagation delay network like satellite network.
- (D) There is no contention in a CSMA/CD network.



Topic : Networking Devices



- Physical Layer : Layer - 1 Device
[Repeater and Hub] → Not store and Forward
 - Data Link Layer : Layer - 2 Device
[Bridge / Switch]
 - Network Layer : Layer - 3 Device
[Router]
 - Application Layer : Layer - 7 Device
[Gateway]
- Store and Forward



Topic : Repeater



- Physical Layer [Layer - 1 Device]
- Two port device
- Signal Regenerator
[Regenerate the signal into output link]
- One bit delay
[Bit by bit forwarding] $\Rightarrow$ No store and Forward.



Topic : Repeater



End-to-end delay for one packet from Host A to Host B

$$= \underline{t_x} + \underline{t_{p1}} + \underbrace{[(1 \text{ bit}) / \text{Bandwidth}]} + \underline{t_{p2}}$$

$$= \boxed{t_x + (t_{p1} + t_{p2})}$$



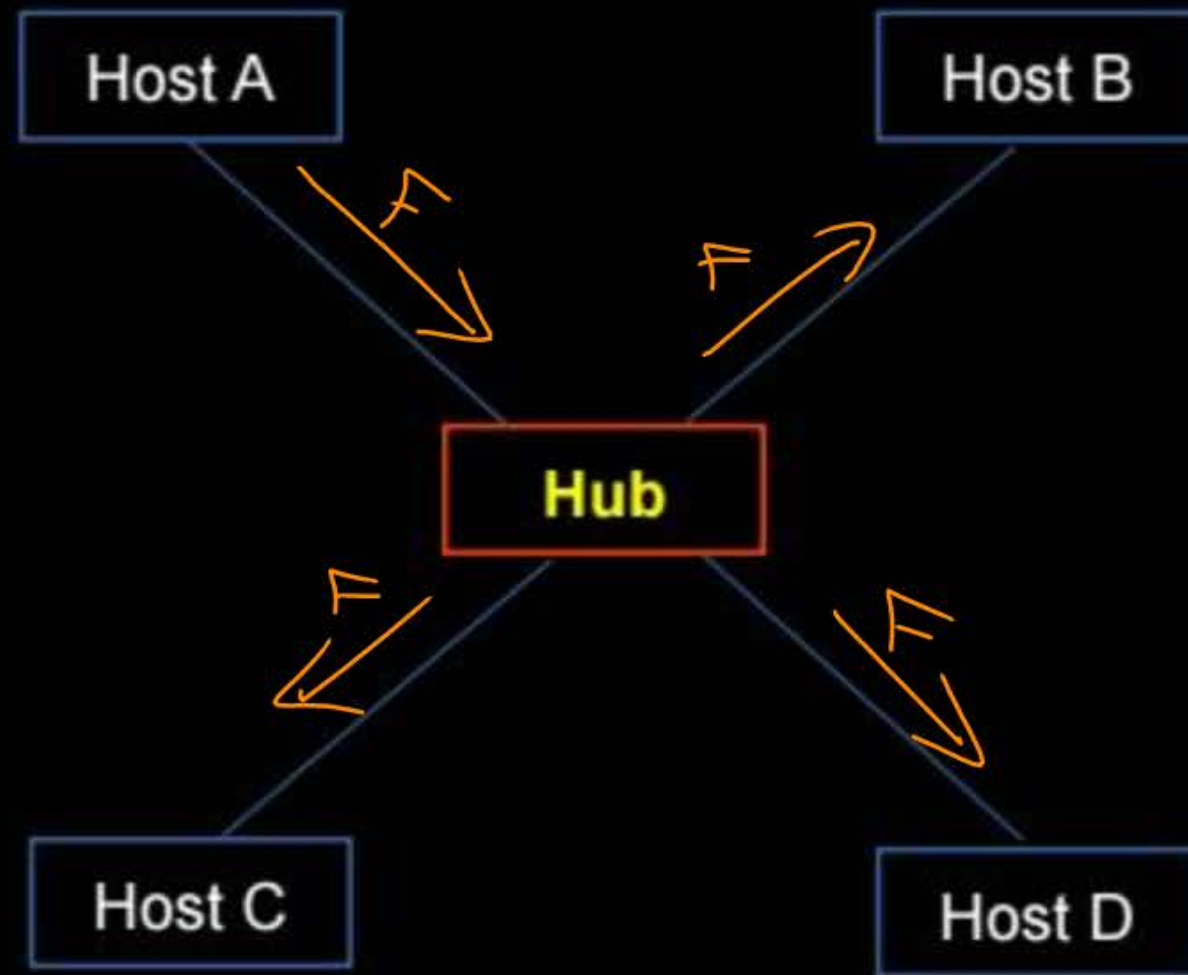
Topic : Hub



- Physical Layer [Layer - 1 Device]
- Multipoint repeater
- Regenerate the signal into all connected links, except incoming link
[Broadcasting device]
- One bit delay [Bit by bit forwarding]

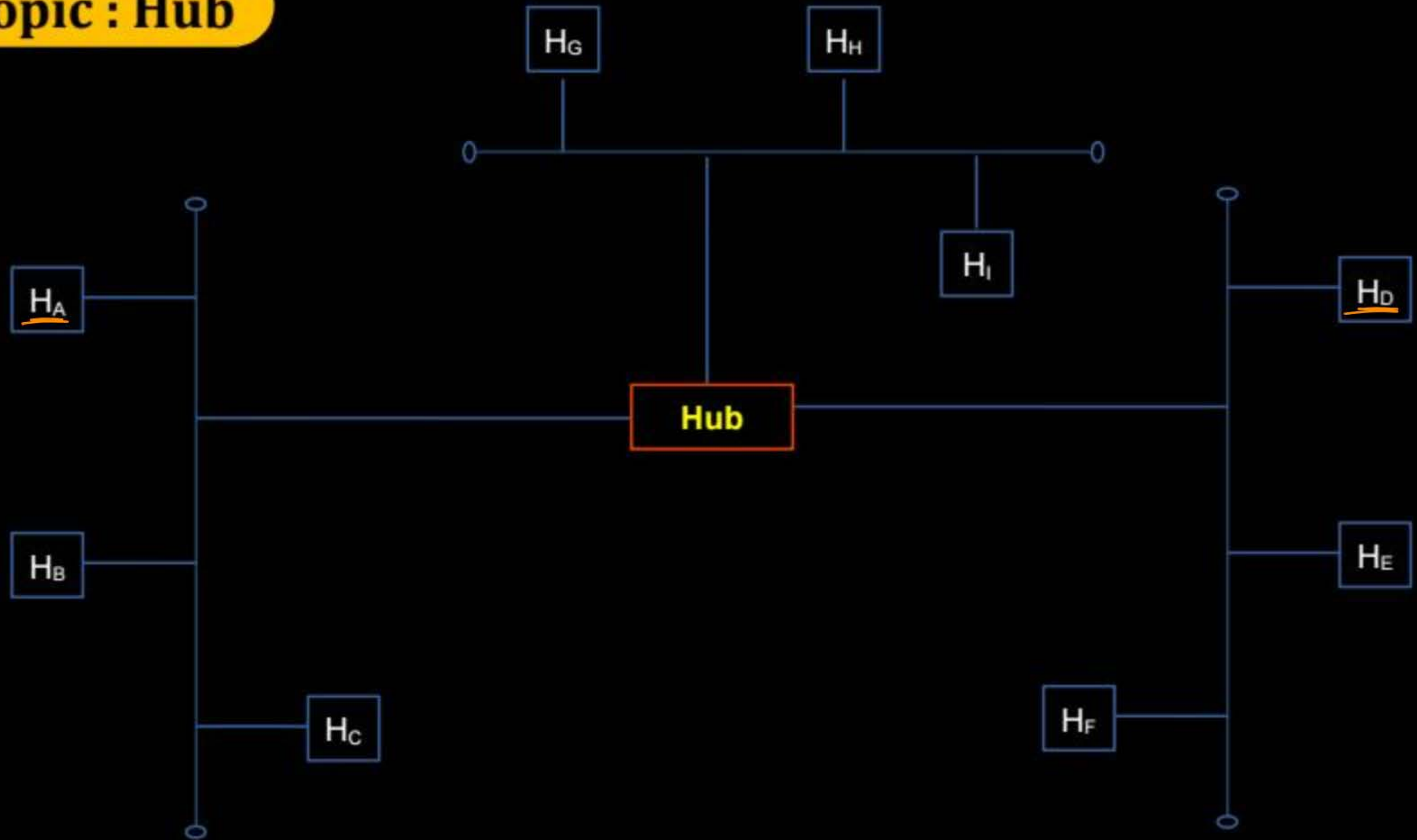


Topic : Hub





Topic : Hub



Source MAC Address = Host A MAC Address

CASE I :

Destination MAC Address = Host C MAC Address

CASE II :

Destination MAC Address = Host D MAC Address

CASE III :

Destination MAC Address = Broadcast MAC Address [FF:FF:FF:FF:FF:FF]

(48 bits are one)



Topic : Hub



→ Not a broadcast domain separator
[Entire LAN has same broadcast domain]

→ Not collision domain separator
[Entire LAN has same collision domain]

Collision domain : Place where collision can occur



Topic : Switch



→ Data Link Layer [Layer - 2 Device]

→ Old Name : Bridge
[Multiport Bridge]

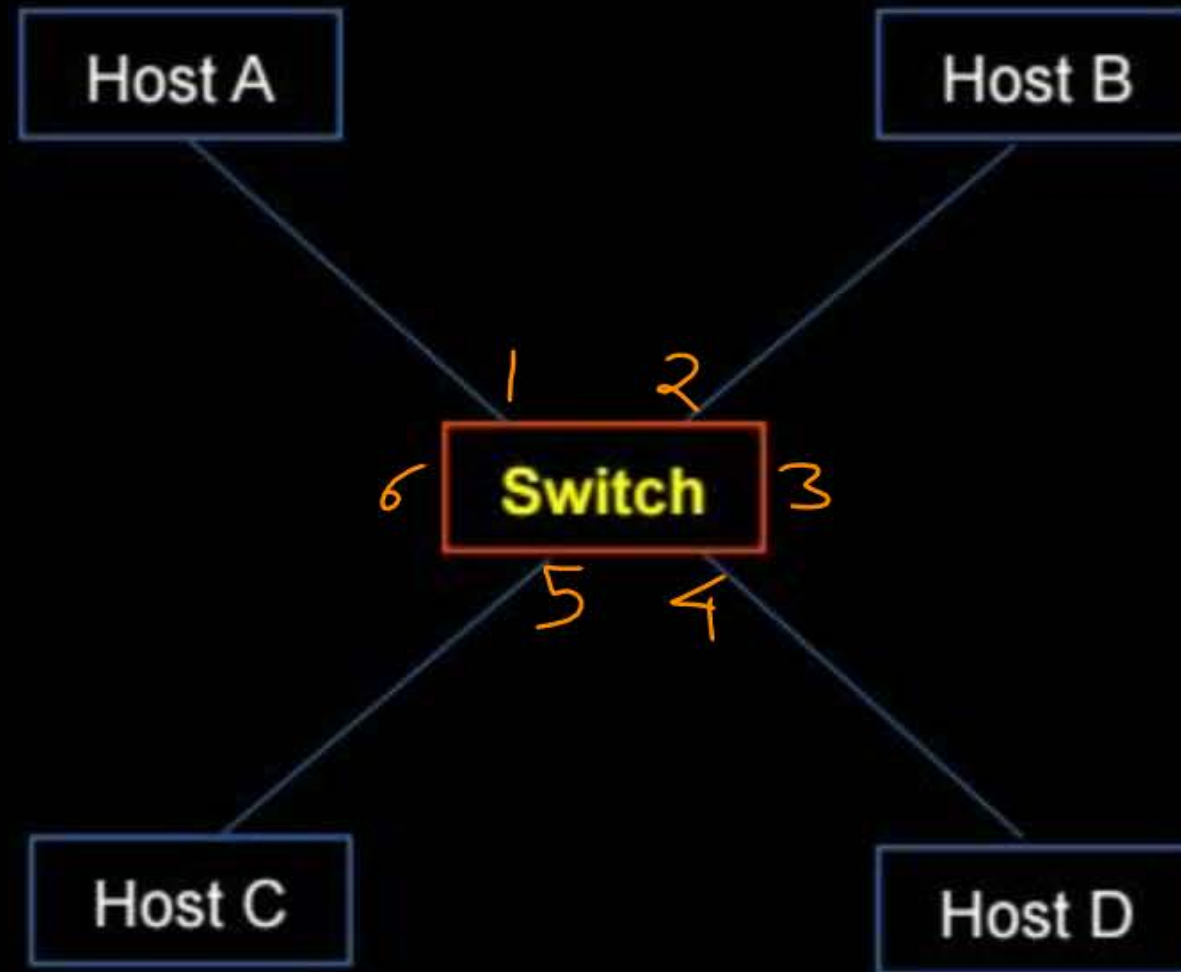
→ Store and Forward → [Frame]
[Store, Process and Forward]

→ Switching Table [Forwarding table] → (Bridging Table)
[Forwarding based on Destination MAC Address]

Transparent switch
[not counted as
network node]



Topic : Switch





Topic : Switch



Two types of switch :

1. Static

- > Non-adaptive
- > Manual update

2. Dynamic

- > Adaptive ✓
- > Self learning
- > Plug-and-play



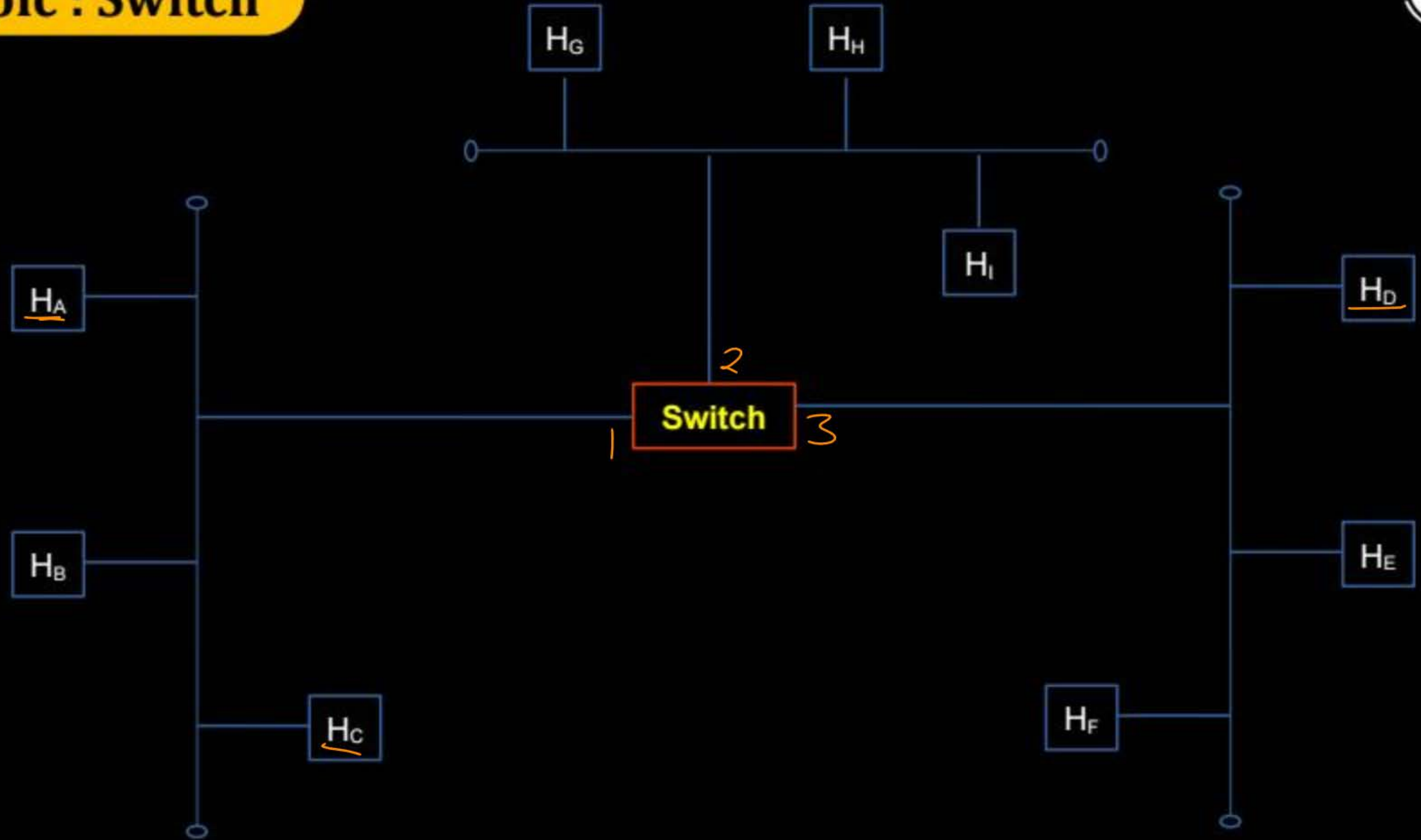
Topic : Switching Table



MAC Address	Interface	[TTL]
H _A [MAC] ✓	1	30 sec
H _D [MAC]	4	30 sec
→ [FF:FF:FF:FF:FF:FF]	[Broadcast]	[Infinite]
[Default]	[Broadcast]	[Infinite]



Topic : Switch



Source MAC Address = Host A MAC Address

CASE I :

Destination MAC Address = Host C MAC Address

⇒ Switch
Discard the
frame
source and dest. MAC
are connected with
same port

CASE II :

Destination MAC Address = Host D MAC Address

CASE III :

Destination MAC Address = Broadcast MAC Address [FF:FF:FF:FF:FF:FF]



Topic : Switch



→ Not a broadcast domain separator
[Entire LAN has same broadcast domain]

→ Collision domain separator
[Different LAN segment has different collision domain]

Collision domain : Place where collision can occur

#Q. Which of the following statements is FALSE regarding a bridge?

- (A) Bridge is a layer 2 device
- (B) Bridge reduces collision domain
- (C) Bridge is used to connect two or more LAN segments
- (D) Bridge reduces broadcast domain

[GATE 2005]

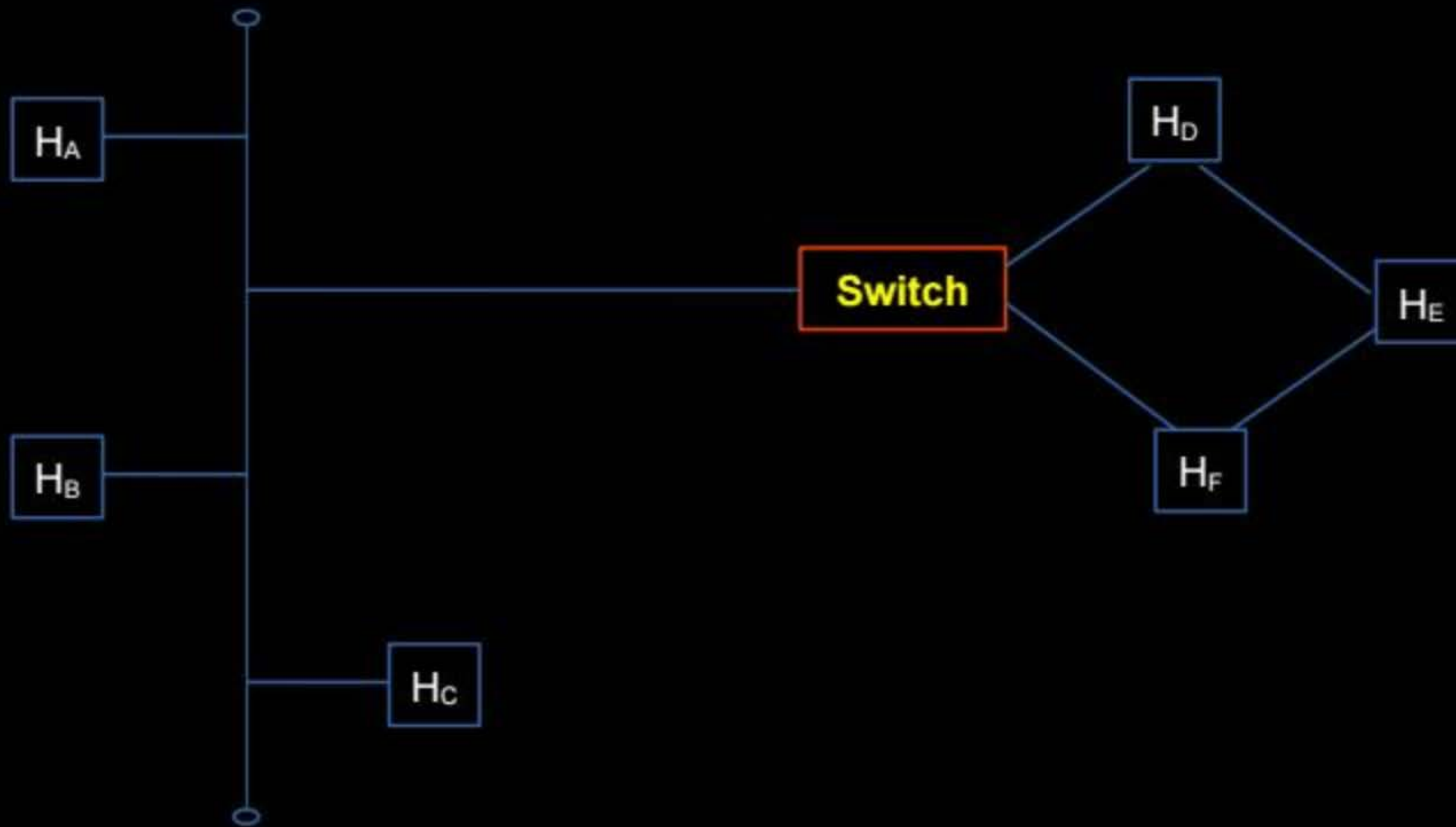
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H.W.*



Topic : Switch



→ To connect different type of LAN segments minimum “Layer-2 device” required.

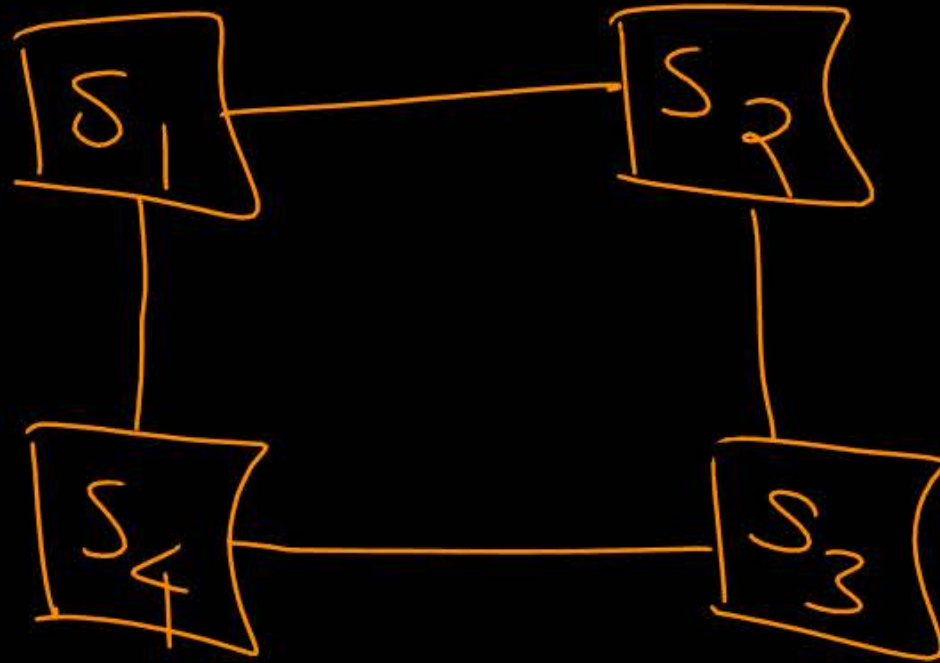




Topic : STP



- Multiple switches inside a network may form circuit (loop)
- To avoid indefinite traversing of frames switches uses "Spanning Tree Protocol (STP)"



#Q. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

- (A) For shortest path routing between LANs
- (B) For avoiding loops in the routing paths
- (C) For fault tolerance
- (D) For minimizing collisions

[GATE 2005]

IIT-B
H.W.



Topic : Router



→ Broadcast domain separator
[Different subnets have different broadcast domain]

→ Collision domain separator
[Different subnets have different collision domain]

Collision domain : Place where collision can occur



2 mins Summary



Topic

Ethernet ✓

Topic

Back-off Time ✓

Topic

Ethernet Bridging ✓



THANK - YOU

